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# HMIS Driven Interactive NMEP Dashboard

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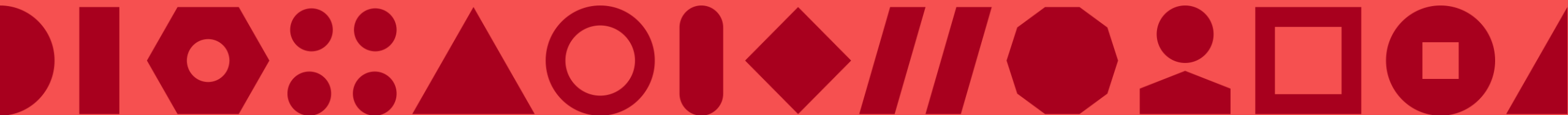
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# Background



# Background: Malaria in Ethiopia

- **~75% of Ethiopia's population lives in malaria-risk areas**, making malaria an important national public health concern.
- **Malaria cases have resurged since 2022**, following substantial progress in malaria control and elimination.
- Malaria transmission varies substantially by **geography and season**, requiring continuous monitoring of where and when the burden is changing.
- This evolving landscape reinforces the need for **timely surveillance to support program planning and response**.

# Malaria Surveillance in Ethiopia

- Ethiopia has two major national health data systems: HMIS and PHEM.

| PHEM                        | HMIS                               |
|-----------------------------|------------------------------------|
| • Weekly (use WHO EPI Week) | • Monthly (use Ethiopian Calendar) |
| • Early warning             | • Routine monitoring               |
| • Rapid response            | • Trends & performance             |

- HMIS provides a primary evidence base for routine monitoring and management of the National Malaria Elimination Program.
- Malaria information in HMIS is captured through two reporting streams: **Service reporting** and **Disease reporting**.

HMIS provides a primary evidence base for routine malaria monitoring and program management.

# HMIS Data Elements

## Service

- Malaria testing and positive results
- Malaria diagnoses
- LLIN distribution/coverage indicators
- Elimination-related investigation and notification indicators
- Other service delivery and program indicators

## Disease

### Malaria cases by diagnosis/species

- *P. falciparum*
- *P. vivax*
- *P. ovale*
- *P. malariae*
- Mixed malaria
- Other/unspecified malaria
- Malaria without parasitological confirmation

**2018 → mid-2023**

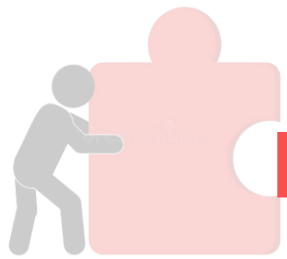
- ✓ ICD-10 based classification

**Mid-2023 → present**

- ✓ ICD-11 based classification

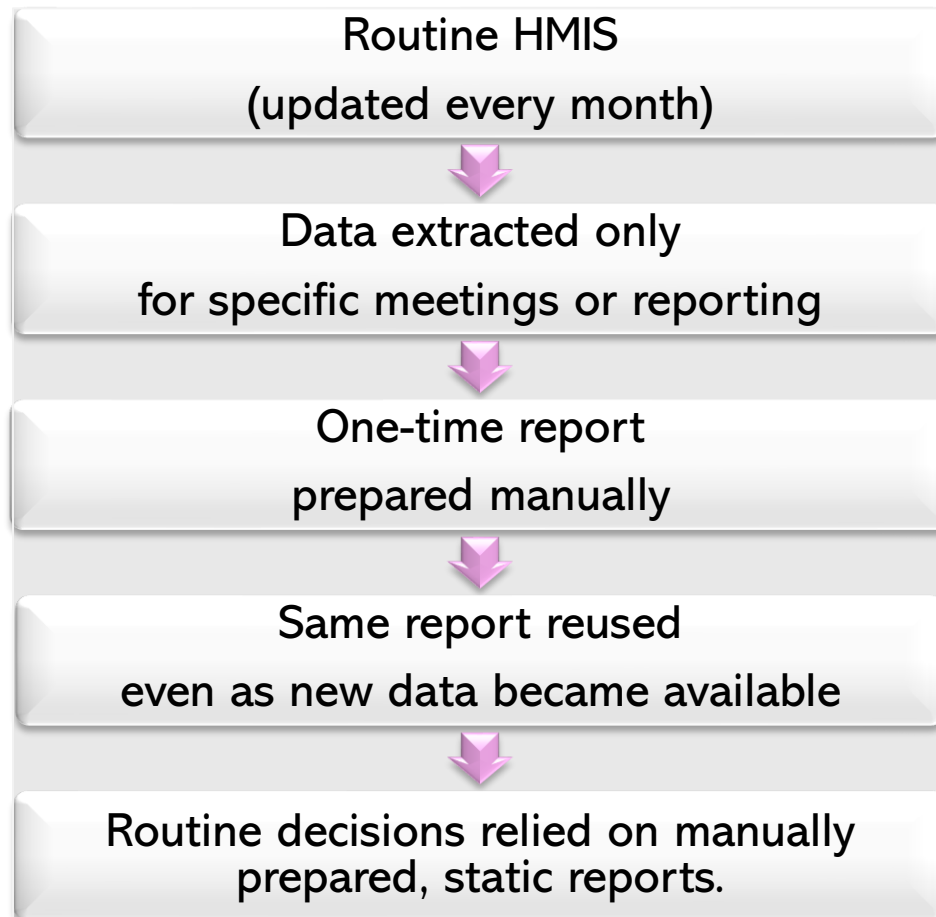
# Rationale





# Rationale

*The current process....*



## Routine malaria data were available, But timely decision support was limited

*Opportunities....*



### Repeated analytical effort

Analyses recreated for different needs



### Limited routine updating

Static outputs require updating as new data become available



### Limited on-demand access

Quick access to key indicators was not routinely available



The aim was to make existing data more readily accessible and actionable for routine program decision-making.



# Objectives



# Objectives



- To develop an interactive HMIS-driven malaria dashboard to strengthen routine monitoring and evidence-based decision-making.

## **1. Enable timely monitoring**

Provide interactive access to key malaria trends, geographic patterns, and program indicators.

## **2. Support data-driven decisions**

Facilitate evidence-based monitoring and decision-making at national and sub-national levels.

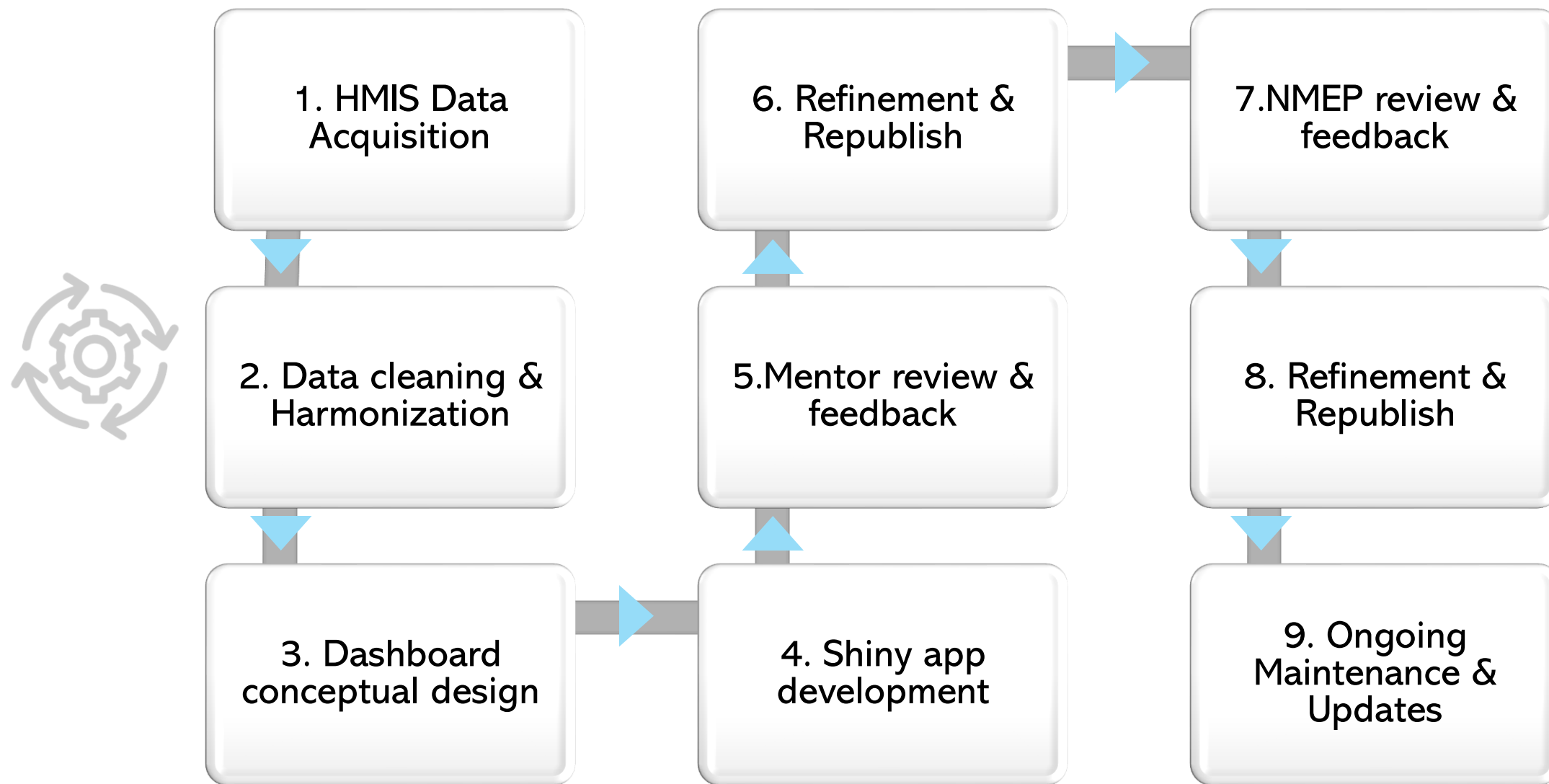
## **3. Establish a reproducible workflow**

Develop standardized data-processing scripts and documentation to support routine dashboard updates and maintenance.

# Workflow



# Workflow overview





# Step 1- Data Acquisition

HMIS/DHIS2 data were acquired as two separate reporting streams

- **Service report:** extracted using default reporting dimensions
- **Disease report:** extracted using default dimensions + **department** and **outcome**
- 9 core HMIS malaria data elements were used (7 disease + 2 service)

| Disease Data Elements                                                                                                                                                                                                                                                                                                       | Service Data Elements                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>7 malaria case categories</b> <ul style="list-style-type: none"><li>• <i>P. falciparum</i></li><li>• <i>P. vivax</i></li><li>• <i>P. ovale</i></li><li>• <i>P. malariae</i></li><li>• Mixed malaria</li><li>• Other parasitologically confirmed malaria</li><li>• Malaria without parasitological confirmation</li></ul> | <b>2 core indicators</b> <ul style="list-style-type: none"><li>• Malaria tests performed</li><li>• Positive malaria tests</li></ul>                  |
| <b>Derived indicators</b> <ul style="list-style-type: none"><li>• Test Positivity Rate (TPR)</li><li>• Malaria incidence per 1,000 population</li><li>• Malaria mortality rate per 100,000 population</li></ul>                                                                                                             | <b>Data quality dimensions</b> <ul style="list-style-type: none"><li>• Reporting <b>completeness</b></li><li>• Reporting <b>timeliness</b></li></ul> |



## Step 2.1- Data Integration & Structural Standardization

### Data Integration

- Restructured service and disease datasets into **long format**
- Standardized both datasets to a **common analytical structure**
- Combined the two reporting streams into a **unified dataset**

### Structural Standardization

- Standardized variable names using `janitor::clean_names()`
- Added common dimensions across datasets where required
- Preserved the distinction between **service and disease reporting streams**

```
19 # 2. changing disease data to long format
20 df_species_long <- df_species |>
21   pivot_longer(cols = contains("ESV"),
22               names_to = "data_type",
23               values_to = "value")

37 # changing the service data to long format
38 df_service_long <- df_service |>
39   pivot_longer(cols = c("MAL_Slides or RDT Positive",
40                       "MAL_Slides or RDT performed for malaria diagnosis"),
41               names_to = "data_type",
42               values_to = "value")

49 # add dept and outcome dimensions to service data
50 df_service_long <- df_service_long |>
51   mutate(department = NA_character_,
52          outcome = NA_character_)

54 # create a unified dataset
55 combined_hmis <- bind_rows(df_species_long, df_service_long)
```



## Step 2.2- Data Harmonization & Enrichment

### Data Harmonization

- Removed redundant text and trailing spaces
- Standardized inconsistent formatting
- Converted Ethiopian Calendar reporting periods to the **Gregorian calendar**
- Standardized geographic identifiers and names
- Resolved inconsistencies across reporting locations

### Data Enrichment

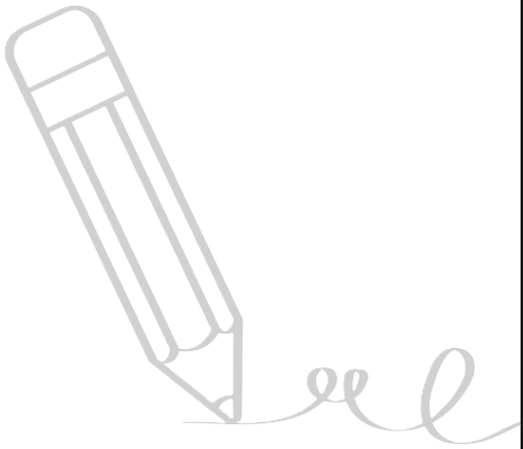
- Integrated **projected population estimates** with HMIS data
- Enabled calculation of **incidence and mortality rates**

```
13 # standardization
14 combined_hmis <- combined_hmis |>
15   janitor::clean_names() |>
16   mutate(region= gsub("Region", "", region),
17          region= gsub("region", "", region),
18          region= gsub("City Administration", "", region),
19          region= gsub("al Health Bureau", "", region),
20          region= gsub("Ethiopian", "", region),
21          region= gsub("Ethiopia", "", region),
22          region= gsub(" ", " ", region),
23          region= str_trim(region)) |>

126 # Apply calendar conversion to combined_hmis
127 combined_hmis <- combined_hmis %>%
128   mutate(
129     eth_month = word(period, 1),
130     eth_year  = as.integer(word(period, -1)),
131     #eth_month = str_extract(period, "[A-Za-z]+"),
132     #eth_year  = as.integer(str_extract(period, "[0-9]+")),
133     eth_month_num = match(eth_month, eth_months),
134     greg_date = eth_to_greg(eth_year, eth_month_num, 1))

80 #joining population data to the hmis
81 hmis_pop_data_integrated <- hmis_agg |>
82   left_join(
83     geo_lookup_long_agg, by= c("region", "zone",
84                               "greg_year" = "year")
85   )
```

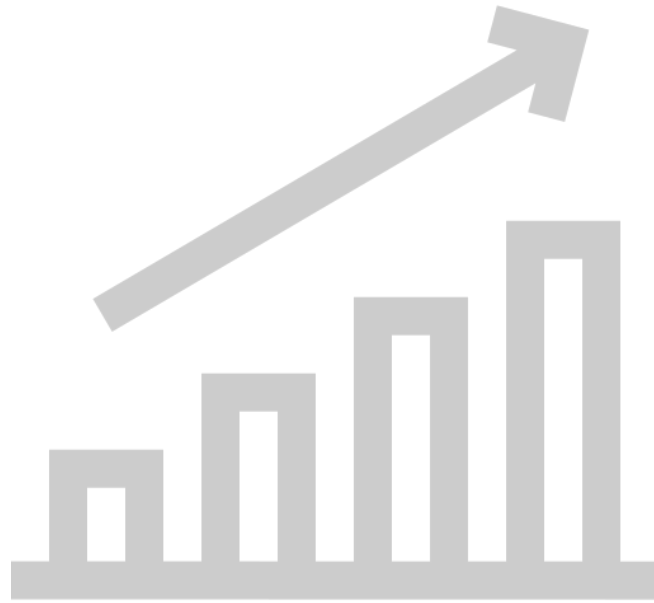
## Step 3: Dashboard Conceptual Design



- **Define dashboard content**
  - ✓ Identified key filters, indicators, metrics, and visualizations
  - ✓ Determined how information should be grouped and presented
- **Design for interpretation**
  - ✓ Selected visualization types appropriate to each indicator
  - ✓ Prioritized a simple, user-focused layout
- **Prototype and refine**
  - ✓ Sketched initial layouts before development
  - ✓ Iteratively revised the design based on review and usability considerations

Program needs → Content → Layout → Visualization → Refinement





## Step 4 - 9: Dashboard Evolution

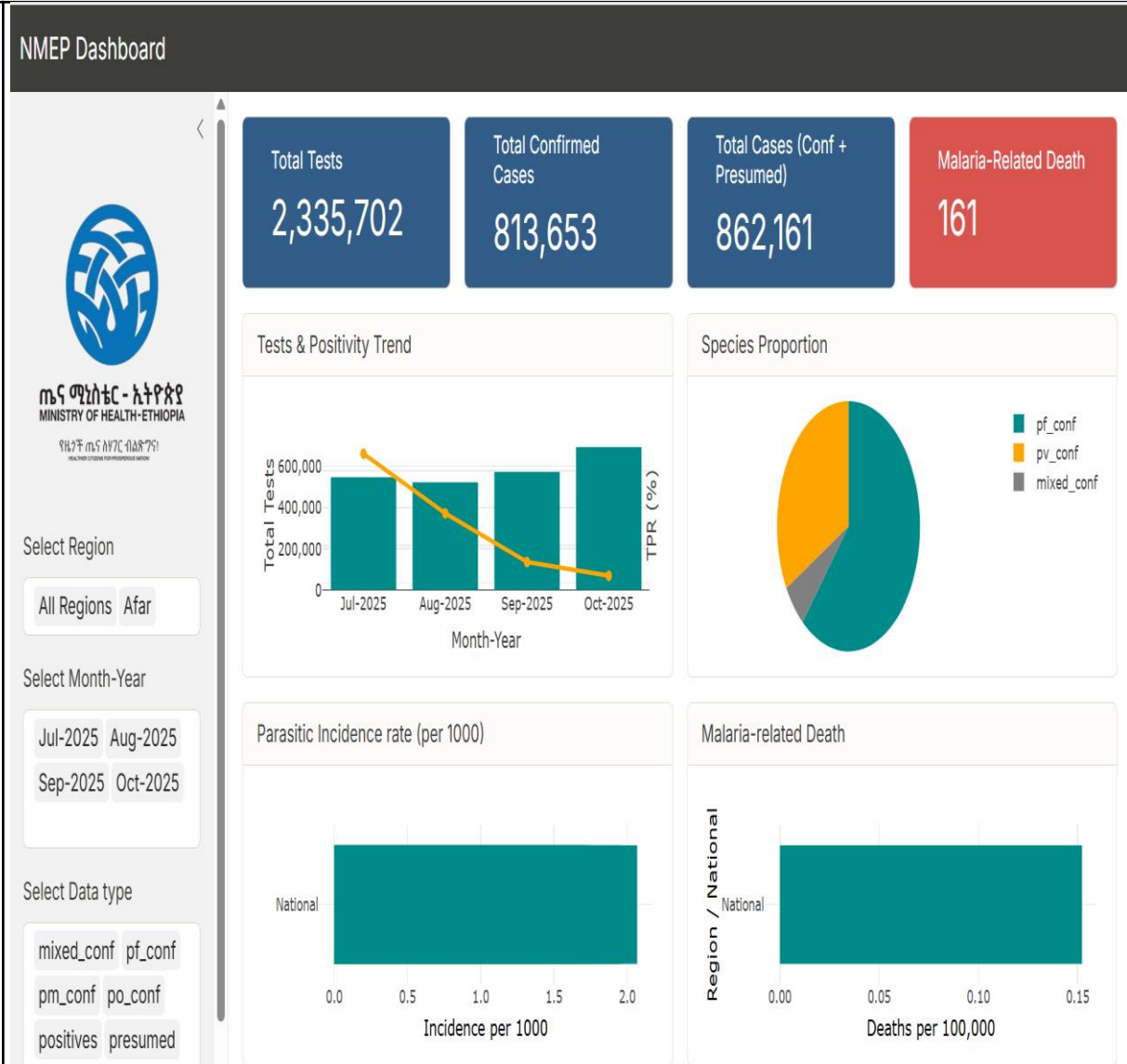
Prototype → Mentor feedback → Refinement → NMEP feedback → Refinement → Current version



# Version 1: Initial Prototype

## Key Lessons from the Initial Prototype

- **Limited geographic granularity**
  - ✓ National and regional views only
- **Conflicting filter selections**
  - ✓ Filter structure could lead to confusing combinations
- **Unclear aggregation**
  - ✓ Indicator values lacked clear definitions of how they were aggregated
- **Unavailable data not clearly handled**
  - ✓ Some dashboard elements appeared blank when data were unavailable
- **Limited temporal comparison**
  - ✓ Few options for exploring and comparing trends over time
- **Limited interpretation support**
  - ✓ Indicator definitions and contextual notes were not yet included





# Intermediate Version: Expanded Functionality

## Key enhancements introduced

- **Expanded geographic analysis**
  - ✓ Added zonal-level views
- **Spatial visualization**
  - ✓ Introduced geographic mapping
- **Alternative indicator views**
  - ✓ Added toggles for TPR and species composition
- **Refined filtering**
  - ✓ Improved filter structure and added age-group selection
- **Remaining refinement need**
  - ✓ Aggregation of API and mortality indicators required further improvement



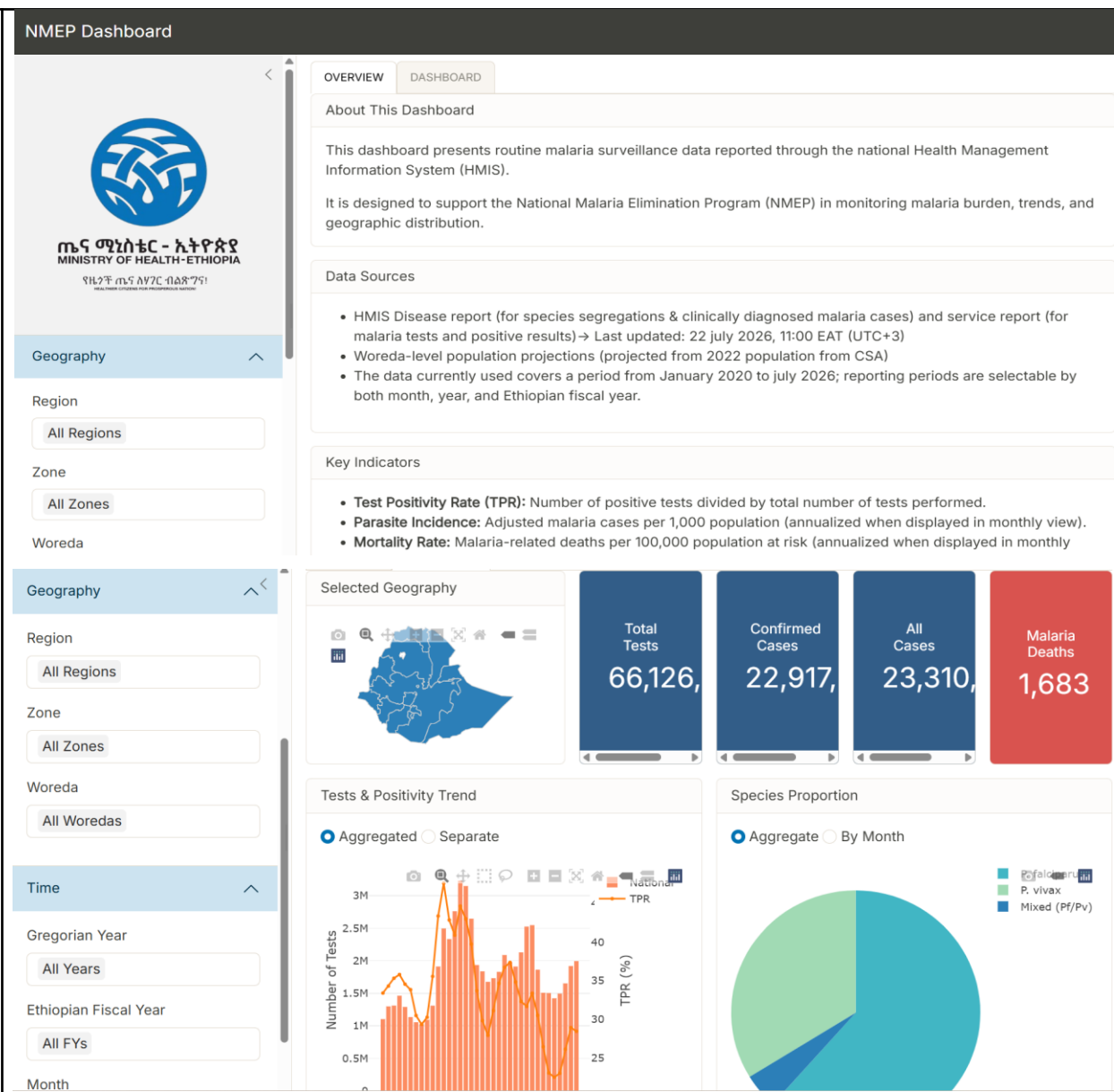


# Current version: Program-Focused Dashboard

## Major refinements

- **Program overview introduced**
  - ✓ Provides a high-level summary of malaria patterns and indicators
- **Geographic analysis expanded**
  - ✓ Interactive mapping extended to **district level**
- **Temporal analysis improved**
  - ✓ Separate **Gregorian year/month** and **Ethiopian Fiscal Year (EFY)** filters
- **Filters optimized based on data assessment**
  - ✓ Facility-type filtering removed from core indicators where incomplete classification could affect values
- **Indicator exploration expanded**
  - ✓ Time-series views added for **API** and **mortality**
- **Data quality monitoring incorporated**
  - ✓ Interactive completeness and timeliness views, including facility-type breakdowns

*Data scope refined: Current version uses HMIS data from **July 2023 onward (EFY 2016+)** to maintain application responsiveness following expansion of geographic granularity and functionality.*



# About the Product



# Product Description

## 1. Data-processing backend

- **R scripts** for cleaning, harmonization, transformation, and data preparation
- Produces the **dashboard-ready dataset**

## 2. Shiny application

- R Shiny code that connects the processed data to the dashboard
- Defines indicators, calculations, visualizations, filters, and application logic

## 3. Interactive frontend

- Web-based interface for exploring malaria information
- Users can filter by:
  - **Geography**
  - **Time**
  - **Facility type\***
- Displays KPIs, trends, maps, and analytical visualizations

## 4. Supporting documentation

### SOPs + User Guide + Data-processing documentation

- *Support routine updating, use, maintenance, and knowledge transfer.*

*Facility-type filtering is applied to data quality views because of incomplete facility classification in HMIS.*

[https://enku-demssie.shinyapps.io/Ethiopian\\_NMEP\\_dashboard/](https://enku-demssie.shinyapps.io/Ethiopian_NMEP_dashboard/)

<https://preelon.github.io/HMIS-pre-processing-/>

# Product Scope

## Data

- National HMIS / DHIS2
- Projected population estimates

## Time

- July 2023 onward (EFY 2016+)
- Monthly trends
- Annual trends
- Seasonal patterns

## Geography

- National
- Regional
- Zonal
- District

## Key indicators

- Tests performed
- Confirmed cases
- Presumed cases
- Test positivity rate
- Incidence
- Malaria deaths/Mortality rates
- Species distribution

# Benefits to the MOH





# How the Dashboard Supports NMEP

## 1. Strengthens routine malaria monitoring

- Provides readily accessible views of trends, **geographic patterns**, and **key indicators** across time and administrative levels.

## 2. Streamlines review and decision-making

- Provides **standardized visualizations and summary metrics** for program reviews, planning, leadership briefings, and routine discussions.

## 3. Supports targeted program management

- Provides evidence to inform **supportive supervision, prioritization of technical support, and sub-national planning**.

## 4. Facilitates identification of emerging patterns

- Helps users identify **increasing trends, geographic disparities, and unusual patterns** that may warrant further investigation or program attention.

The dashboard makes routinely available HMIS data easier to access, explore, interpret, and use for program decision-making.

# Challenges and Mitigation Strategies



# Potential Challenges and Mitigation Strategies

| Potential challenge                            | Potential impact                                                   | Mitigation strategy                                                                                                  |
|------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Limited or delayed access to updated HMIS data | Delayed dashboard updates                                          | Establish a routine process for obtaining updated HMIS extracts and maintain coordination with relevant MoH offices. |
| HMIS data-quality issues                       | Inaccurate or misleading outputs                                   | Apply data-quality checks and validation rules; flag anomalies for review before publication.                        |
| Technical or hosting issues                    | Dashboard downtime or reduced functionality                        | Maintain backup application files and scripts; test deployments and engage IT support when needed.                   |
| Changes to HMIS structure or indicators        | Processing workflow or dashboard functions may become incompatible | Maintain modular, adaptable scripts and update the workflow when HMIS structures change.                             |
| Limited ongoing technical capacity             | Difficulty maintaining and updating the dashboard                  | Provide hands-on knowledge transfer and maintain SOPs, user guides, and technical documentation.                     |

# Lessons learned



# Lesson learned

## 1. Build iteratively, not once

- Dashboard development required repeated cycles of development, testing, feedback, refinement, and republishing.

## 2. Understand the data before visualizing it

- A sound understanding of HMIS structures, reporting processes, indicator definitions, and data limitations is essential for meaningful visualization.

## 3. Start with the program's questions

- Understanding how the program monitors malaria and what information teams need should guide the choice of indicators, filters, and views.

## 4. Test features against the data

- A technically functional feature may not always be appropriate. Data-quality assessment and validation should inform dashboard design decisions.

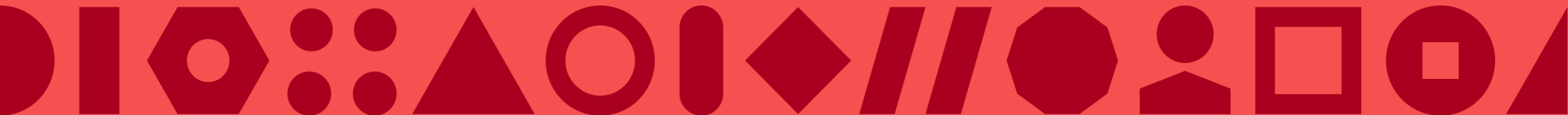
## 5. Seek feedback and learn from others

- Feedback from **mentors and NMEP colleagues** helped identify issues that were not always apparent during development and improved the relevance and usability of the product.

## 6. Design for sustainability from the beginning

- Reproducible scripts, SOPs, user guidance, and knowledge transfer should be developed alongside the product rather than added at the end.

# Sustainability plan



# Sustainability plan

## 1. Institutional Ownership

- Transfer source code, **data-processing scripts, and deployment materials** to NMEP
- Host the application under an **institutional/program-owned account**
- Assign responsibility for **routine maintenance and updates**

## 2. Capacity Transfer

- Provide **hands-on training and focused mentorship** to designated NMEP staff
- Build practical skills in **data processing, visualization, and dashboard maintenance**
- Gradually transition routine updating and maintenance to program staff

## 3. Standardized & Reproducible Workflow

- Document the **monthly data preparation and update process**
- Maintain **SOPs and user guides** for data preparation, dashboard updating, and use
- Use reproducible scripts to support **consistent and repeatable updates**

## 4. Continued Support & Improvement

- Maintain a **monthly update cycle** aligned with HMIS data availability
- Continue focused support during the **transition period**
- Review and adapt the dashboard as **program needs, indicators, and HMIS structures evolve**

